

Problem Set: Numerical Methods

Geophysics 3A

1 Background

Cooling through oceanic seafloor is relatively well-characterized by a plate cooling model. Heat flow patterns at the surface are more complicated as hydrothermal circulation redistributes heat in the upper oceanic crust. Sediments covering the oceanic crust cut off ventilated flow directly between the igneous crust and the oceans. However, the igneous basement has sufficiently high permeability that fluids can continue to circulate beneath the sediments.

In this practical, we will use a finite difference solution to the heat equation to model heat flow through sediments and to the surface.

Modeling heat flow through sediments is complicated—not because the physics is complex—but because the geometry is irregular. The depth to the sediment–basement interface varies laterally, and sediment compaction alters the physical properties of the sediments with depth.

Important assumption: Thermal conductivity and heat production vary **only with depth** due to compaction, while sediment thickness varies laterally. This combination produces a fully two-dimensional temperature and heat flow field.

Because of these complexities, a numerical approach is required rather than an analytic solution.

2 Finite Difference Formulation

We will construct a 2-D steady-state solution to the heat equation. In the numerical grid:

- Temperatures and depth are defined at **nodes**
- Thermal conductivity and heat production are defined at **cell centers**

Follow the steps below carefully.

1. Governing equation

Task: Write down the partial differential equation governing steady-state heat conduction in two dimensions with spatially variable thermal conductivity k and heat production A .

Hint: Combine conservation of energy with Fourier's law.

2. Boundary conditions

The temperatures at the ocean–sediment interface and the sediment–basement interface are fixed. However, the side boundaries require special treatment.

(a) Top and bottom boundaries

Task: Write expressions for:

- the surface temperature
- the basal temperature

Note: The basal temperature may vary laterally.

(b) **Side boundaries (insulating)**

We assume no heat can flow across the vertical sides of the model.

Task:

- Write the mathematical condition for zero heat flux
- Express this condition in finite difference form

3. Finite difference equation

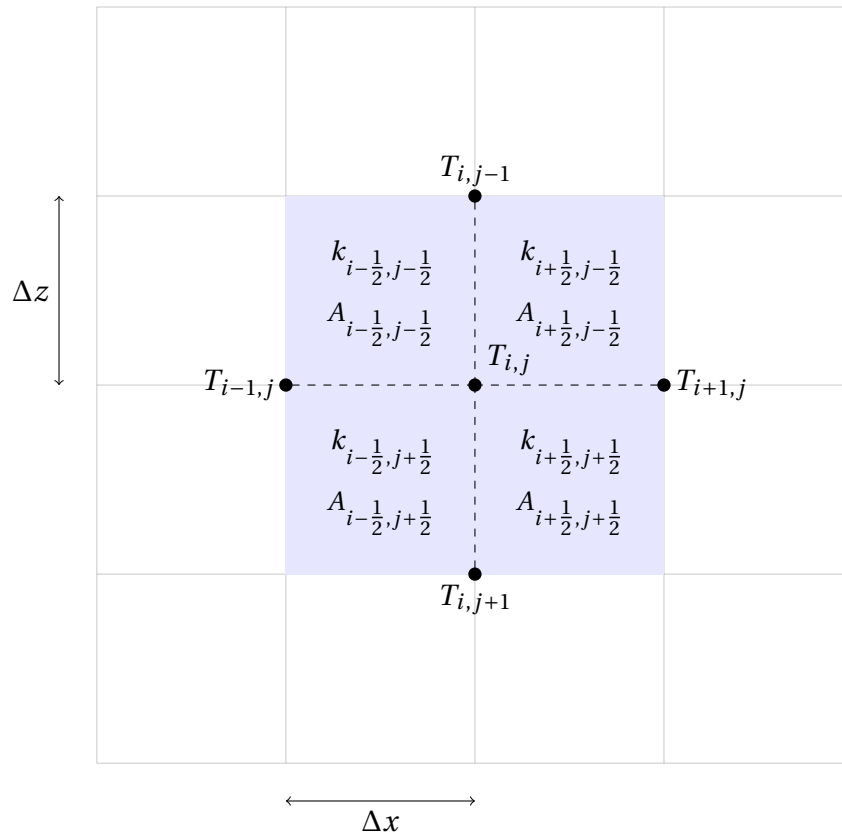


Figure 1: Stencil for the 2-D thermal problem. The temperatures are *node centered* (defined at the nodes, circles). The thermal properties are *cell centered* (defined at the center of the cells, rectangles). The cell positions are defined in terms of (i, j) as the (column, row) positions associated with (x, z) .

Task: Derive the finite difference equation for the temperature at node (i, j) using a central difference approximation.

Important:

- Write the equation in terms of heat fluxes entering and leaving the node
- Use conductivities defined at the **interfaces** (up, down, left, right)

Your answer should clearly show:

- contributions from all four neighbouring nodes
- the role of heat production at the node

4. Heat production at nodes

Heat production is defined at cell centres, but the finite difference equation requires values at nodes.

Task:

- (a) Derive an expression for heat production at node (i, j) using the arithmetic mean of the four surrounding cells.
- (b) **Then simplify:** Assume heat production varies only with depth (i.e. $A = A(z)$). What does your expression reduce to?

5. Thermal conductivity in four directions

Thermal conductivity varies with depth due to compaction.

Task: Write expressions for conductivity in the four directions (up, down, left, right) from node (i, j) .

Hint: Use the arithmetic mean of adjacent cell values in the direction of heat flow.

6. Porosity-controlled properties

Sediment compaction reduces porosity with depth according to:

$$\phi = \phi_0 \exp(-z/\lambda) \quad (1)$$

Tasks:

- (a) Thermal conductivity depends on both the sediment matrix and pore fluid. Using a harmonic mean, show that:

$$k = \left(\frac{1-\phi}{k_m} + \frac{\phi}{k_w} \right)^{-1}$$

- (b) Heat production is associated with the solid matrix only. Show that:

$$A = A_0(1 - \phi)$$

- (c) **Explain qualitatively:**

- How does conductivity vary with depth?
- How does heat production vary with depth?

7. Surface heat flow

Task: Derive an expression for vertical surface heat flow using your finite difference solution.

Hint: Use the temperature gradient between the uppermost nodes.

8. Conceptual question (important)

Question:

If thermal conductivity k and heat production A vary only with depth (and not laterally), what causes the temperature and heat flow field to be two-dimensional?

Answer clearly in words.

Hint: Think about the role of geometry.

9. Sketching exercise

Draw a cross section across a buried bathymetric high.

Tasks:

- Draw two cases:

- (a) 1-D (uniform sediment thickness)
- (b) 2-D (varying sediment thickness)
- On each sketch, show:
 - isotherms
 - heat flow directions

3 Computational Exercise

After completing the formulation, run `iberia2d.py`.

The data are from heat flow measurements collected on the Iberian Abyssal Plain. Sediment thickness is derived from seismic data, and thermal properties are estimated from measurements and core samples.

The goal is to develop a 2-D thermal model that matches the observations.

10. Model comparison

The blue model is 1-D and the red model is 2-D.

Tasks:

- Describe how well each model fits the observed data
- Identify where each model performs well or poorly

11. 3-D effects

Task: Describe how heat flow over a 2-D ridge differs from that over a buried seamount (a 3-D feature).

Answer in words:

- How do heat flow patterns differ?
- Would including 3-D effects significantly reduce the misfit?

12. Improving the model

Task: If 3-D geometry does not fully explain the data, suggest ways to improve the model.